

N E X U S Z O N E

Portfolio Companion

Product Catalog · Invention Catalog

EDITION 2026

"Where technology meets craft."

*A complete inventory of shipped products, published
research, and applied-security field work.*

Founded by **game_nexus_zone**

Confidential portfolio companion · Distribute with attribution

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INTRODUCTION

00 — About This Companion

This companion is **not a résumé**. A résumé is a one-page document you read in thirty seconds. This companion is what you read *after* the résumé — when you have decided the products deserve closer inspection.

Every product here follows one design rule: **every claim must be verifiable**. Every number is measurable in a running system. Every architectural choice is defended by a public commit, a working demo, or a peer-review-ready research paper.

“

Products are not judged by the demos they run. They are judged by the demos they refuse to run.

Inside this catalog you will find **twelve shipping products**, **three research papers**, **six Kaggle competitions**, and **nine responsible-disclosure engagements** with organisations including Okta, Magic Labs, LaunchDarkly, Gearset, and Shopify. Each product page states what the product does, what makes it different from its category, the technical signature that gives it edge, and the verified numbers behind its claims.

12+

Shipped products

3

Research papers

9+

Disclosure engagements

51

AI agents in prod

CHAPTER 1 · THE PARENT BRAND

01 — Nexus Zone

Nexus Zone is the umbrella brand under which every product in this catalog ships. The name unifies two heritages:

- **Game Zone** — the original mastery territory: interactive experiences, gamer platforms, community tools.
- **Nexus** — the connection point that links gaming to AI, cybersecurity, and software engineering.

Nexus Zone operates in **eight service lines**, each backed by at least one shipping product:

[Gamer platforms](#)[Digital marketing](#)[Project delivery](#)[Social channels](#)[Cybersecurity](#)[AI solutions](#)[Legacy-code modernization](#)[Enterprise code review](#)

MARKET POSITION

Nexus Zone is the first Arab-founded technology house to combine consumer-facing gaming products (Golden Arena, Waraq Reader) with enterprise-grade engineering tooling (LucidCode, Laundry) under a single Charter. A single customer can build a game, secure a website, modernise legacy code, and receive an Arabic radiology report — from one origin.

Public site: 61465.github.io/game-zone-hub

Founder: game_nexus_zone.

LucidCode

V0.5.0 · PROGRAM VERIFICATION

Version: 0.5.0 **Licence:** Commercial (Nexus Zone 1.0) **Tests:** 330 / 330

Benchmark: F1 = 1.00 on 22 CVE-inspired fixtures

CHAPTER 2 · CODE CONFESSION ENGINE

02 — LucidCode

LucidCode is a developer-experience tool that grants source code cognitive self-awareness. Instead of running static rules over a file, LucidCode makes the code **confess in first person** — and rigorously refutes any confession that cannot be proven.

“
[T1] Suppression @ line 8 — "I confessed: I caught an exception and said nothing.
Every scream is muffled the moment it starts. What am I hiding?"

THE SIGNATURE

THREE-ENGINE ANTI-HALLUCINATION ENSEMBLE

Every confession passes through three independent verifiers before it reaches the user: (1) a deterministic AST re-verifier weighted at 0.90, (2) a sandboxed fuzzer weighted at 0.70, (3) an adversarial LLM devil weighted at 0.40. A Bayesian aggregator produces a calibrated verdict. Hallucinated confessions are logged internally and never displayed.

POSITIONING

- **Semgrep / SonarQube:** static rule match, no reasoning.
- **Copilot:** autocomplete without validation.
- **CriticGPT:** single-model self-critique.
- **LucidCode:** every claim anchored to a real file+line, every fix voted on by three independent engines.

DELIVERABLES

12 CLI sub-commands · VS Code VSIX extension · CodeQL query pack · GitHub App scaffold · arXiv-ready whitepaper · Firecracker sandbox · *Ring* orchestrator (Red-Hat + War-Gaming + Human-Triage roles).

SHIPPING STATUS

Field-validated on **3 real production repositories**. Benchmark suite of 22 CVE-inspired fixtures achieves **F1 = 1.00**. Commercial licence live under Nexus Zone Commercial 1.0. B2B pricing tier ready.

Landing page: 61465.github.io/lucid/landing

Laundry

V0.3.2 · LEGACY-CODE MODERNIZATION

Version: 0.3.2 Tests: 97 / 97 Codebase: 62 Python modules
Battle-tested on: Django (2,974 files), Airflow (9,082 files)

CHAPTER 3 · THE LAUNDRY ROOM

03 — Laundry

Laundry is a legacy-code modernization pipeline that **never merges**. It runs twelve specialised agents on a target repository and produces one signed evidence bundle a human engineer can approve — with a full before/after explanation for every unit of code.



Legacy code goes in. Evidence comes out. The tool never merges.

THE SIGNATURE

IMMUTABLE CHARTER · TEN NON-NEGOTIABLE RULES

At session start Laundry loads a Charter of ten runtime-immutable rules — no auto-merge, evidence before edit, judge never writes code, uncited claims dropped, dead code needs two independent proofs, and more. The Charter is stored in a *MappingProxyType* at runtime; any mutation attempt raises *TypeError*. The Charter protects the tool from the tool's operator.

TWELVE SPECIALISED AGENTS

Each agent has one job. The judge never writes code. The updater refuses to run without a passing baseline. The parity agent isolates every patch in a sandbox before accepting it.

- Indexer
- Security
- Business Rules
- Dead Code
- Tests
- Updater
- Parity
- Architecture
- Explainer ★
- Passport
- Documenter
- Judge

VERIFIED FIELD RESULTS

- **Django** (v5.2 · 2,974 Python files) — analysed in **88 seconds**. Surfaced 289 real secrets, 3,607 business rules, 351 dead-code candidates, zero crashes.
- **Apache Airflow** (main branch · 9,082 files) — analysed in **116 seconds**. 486 real secrets surfaced, 5,000 business rules, 2,000 dead-code candidates, six agent-level contradictions correctly escalated.
- **Flask** (85 files, 5.9 s) and **Requests** (37 files, 3.7 s) — control runs confirming false-positive rate on clean code.

COMMERCIAL READINESS

21 LLM providers wired (Anthropic, OpenAI, Groq, Cerebras, Gemini, Bedrock, Vertex, and more). 15 integration categories (Slack, Sentry, Jira, Linear, etc.) via 138 documented environment keys. SARIF

export lands directly in GitHub / GitLab / Azure DevOps code-scanning surfaces. HMAC signed-override protocol enforces human authorisation before any gate is bypassed.

Interactive live demo: 61465.github.io/lucid/landing/laundry.html (runs against *psf/requests* v2.7.0, May 2015).

NEXUS-AI

V0.7 · 51-AGENT ORCHESTRATION FABRIC

Agents: 51 **Models:** 8 LLMs behind one facade **Skills:** 1,580 curated **Protocol:** MCP JSON-RPC 2.0

CHAPTER 4 · THE 51-AGENT FABRIC

04 — NEXUS-AI

Most teams ship one or three agents. NEXUS-AI ships **fifty-one**, each with a narrow, well-defined role. The value is not in the count — it is in the coordination layer that keeps them useful.

THE SIGNATURE

META-ROUTER WITH 100% TASK CLASSIFICATION ACCURACY

Every request enters through the *Meta-Router*, a rule-based classifier that maps the request to a task type (code_gen · reasoning · web_search · vision · osint · writing ...) and then dispatches to the correct agent and the cheapest capable model. Verified 100% classification accuracy on a 13-request live test suite spanning real production traffic.

VERIFIED RUNNER — THE TRUTH LAYER

Every agent response is filtered through three tiers before it leaves the fabric: (1) Tier-0 schema check, (2) Tier-1 NLI self-check, (3) 13-signature injection detector. Only fully-cleared responses reach the caller.

COMPANY ORCHESTRATOR

For composite tasks, the Company Orchestrator decomposes the request into sub-tasks, dispatches them in parallel across specialised agents, verifies each result, and merges. Functionally, one call yields the output of an engineering department — at LLM latency.

PRODUCTION FOOTPRINT

NEXUS-AI is the shared brain across **midcine** (radiology inference orchestration), **thawani.cc** (customer-response automations), and internal engineering workflows. The Meta-Router prefers free-tier providers (Naraya · Groq · Cerebras), which drives real per-request cost close to zero.

midcine

ARABIC RADIOLOGY AI · RIS/PACS

Target: Hospitals in MENA **Read time:** 24-slice CT in ~69 s **Confidence:** 92%

Output: DICOM SR-signed Arabic report

CHAPTER 5 · RADIOLOGY BEHIND THE FIREWALL

05 — midcine

Most radiology-AI products run in US-hosted clouds. MENA hospitals cannot legally send patient scans there. midcine is designed to run **inside the hospital firewall** — no external network access unless explicitly authorised — and to write reports directly in Arabic, signed as standard DICOM SR objects.

MEASURED PERFORMANCE

- Reads a 24-slice CT into a structured report in **~69 seconds**.
- AI confidence: **92%**.
- Guardian review reproduced an ACR-grade subarachnoid haemorrhage diagnosis.
- Runs on 8 GB RAM minimum, 16 GB recommended.
- Ten micro-services in a single pnpm/turbo monorepo.

THE SIGNATURE

VISION + CLINICAL LLM ENSEMBLE

midcine does not rely on a single model. Its pipeline chains a *Vision Classifier* (lesion class), a *Detection Model* (bounding boxes), a *Segmentation Model* (volume maps), a *Vision-Language* reader (radiological description), and a *Clinical LLM* (final report composition). Each model cross-verifies the previous one; disagreements route to a human reviewer.

OFFLINE-FIRST ROADMAP

Egyptian hospital networks are unreliable. midcine ships with Ollama fallbacks so an entire study can be read without any Internet connection. Optional premium services (TotalSegmentator, external PACS bridging) are opt-in.

Live demo: `ame.tail119ddab.ts.net:8445/reports` (private Tailscale Funnel · synthetic patients).

thawani.cc

WHATSAPP COMMERCE SAAS

Market: Saudi + Gulf **Status:** Live paid tenants **CR:** 3453105300 **VAT:** 311174977200003

CHAPTER 6 · WHATSAPP AS A STOREFRONT

06 — thawani.cc

Every WhatsApp-commerce project hits the same wall: **Meta bans sessions**. Meta's own baseline ban rate is 0.5%. At scale, that means one tenant lost every day.

thawani.cc treats the ban problem as an engineering problem, not an operational one. The response is a **26-layer defence stack** operating in parallel.

THE SIGNATURE

26-LAYER DEFENCE STACK

Stealth Mode v2 · Behavior Analyzer + Gatekeeper (a seven-rule 0–100 scoring model) · Phase-2A humanisation · adaptive rate limiting · reputation tracking · phone-uniqueness enforcement · and twenty other independent layers. Compound effect: the observed ban rate drops from Meta's 0.5% baseline to **0.1%** — a five-fold improvement across live tenants.

SIX VERTICAL MODULES · ONE SHARED KERNEL

Bookings

Dine-in QR

Accommodation

Gaming Topup

Restaurant Reservations

Lamah Calendar

SALLA INTEGRATION

Salla is Saudi Arabia's largest e-commerce platform (55,000+ MENA stores). Phase 1 is live: OAuth 2.0 + Webhooks + product sync + disconnect. Every Salla store can convert to a thawani tenant with a single button.

COMMERCIAL MODEL

SAR 50 / store / month via Meta WhatsApp Cloud API in Tech-Provider mode. Even a 1% Salla penetration yields SAR 27,500 monthly recurring revenue. Six-vertical shared-kernel design keeps unit economics healthy across scale.

ARIA v9.0 / GZP-LLM

SOVEREIGN AI COMPANION

Base: Qwen 2.5-7B Tuning: LoRA r=64 + DPO Training compute: Kaggle GPUs Design goal: one human

CHAPTER 7 · A MODEL FOR ONE HUMAN

07 — ARIA v9.0

Most AI models are "for everyone." ARIA is deliberately **not**. ARIA is designed to grow through an exclusive relationship with a single human — it has memory, a personality, autonomous background cognition, and a curiosity engine that asks real questions to fill its sensory blind spots.

“

Not a chatbot. An intelligence that knows its owner.

THE SIGNATURE

CURIOSITY ENGINE

Every hour, a background "curiosity" pass reviews the day's conversations, identifies gaps in the model's world model, and formulates targeted questions the human is prompted to answer. Over time ARIA **knows its owner better** than any general-purpose assistant.

STRATEGIC VALUE

ARIA is not currently for sale. It is a **life pattern**. Once the internal loop is fully stabilised, the underlying infrastructure will be released as a hosted SaaS so any individual can build their own sovereign companion.

CURRENT STATE

- v9.0 trained and running.
- Real-time gaming overlay live.
- Long-term memory persisted locally.
- Arabic + English bilingual.

NEXUS Vigil™ (Smart Fleet AI)

ABANDONED-VEHICLE INTELLIGENCE

Market: US municipalities **Coverage:** 6 states + 40 cities **Endpoints:** 44 REST

Database: 97 MB live SQLite (WAL)

CHAPTER 8 · EVERY ABANDONED CAR HAS A STORY

08 — NEXUS Vigil™

US municipalities spend millions removing abandoned vehicles. They do not know the vehicle's age, its owner history, or whether it is subject to manufacturer recall. NEXUS Vigil™ resolves all three in a single API call.

THE SIGNATURE

MAPILLARY GRAPH API V4 + NHTSA VPIC / RECALLS

The product fuses two of the strongest open datasets in North America: *Mapillary* (crowdsourced street imagery at scale) and *NHTSA* (vehicle-manufacturer and recall data). Vigil captures a plate, resolves make/year, checks recall status, and opens a case file — in seconds.

PROPRIETARY PRODUCT

Full intellectual-property ownership. Municipalities buy subscription access; live SQLite database auto-updates. Coverage: Massachusetts, Connecticut, Rhode Island, New Hampshire, Vermont, Maine — plus 40 individual cities. Roadmap: EU municipalities.

CHAPTER 9 · THE SECURITY STACK

09 — Applied-Security Products

Four products operate together as a single security fabric:

NEXUS SecOps + NEXUS_AUDIT Arsenal

RED / BLUE / AI / OSINT

32 registered security tools (Amass, Nuclei, Metasploit, CodeQL, garak, PyRIT, Shodan, Censys, and more) plus 4 live threat-intelligence feeds (GreyNoise · Leakix · urlscan · PublicWWW) exposed behind seven REST endpoints. A single command `nexus-audit <target>` wraps 10+ SAST/DAST tools with automatic language detection and language-specific rule packs.

NEXUS Firm

POLICY-GOVERNED ENGINEERING CONTROL PLANE

One architectural rule: *LLM proposes → tools produce evidence → policy decides → agent executes within capability bounds*. Inter-agent handoffs are typed artifacts, never prose. Every state mutation is a validated, logged tool call.

OPA / Rego

MCP Gateway

Firecracker

gVisor

Capability Lattice

Evidence-WORM Memory

SENTINEL Enterprise + sentinel-cli

WINDOWS SECURITY SUITE

Windows SIEM + SOAR + ML in one bundle. For organisations that need endpoint protection on Windows without paying CrowdStrike enterprise pricing. Terminal companion (`sentinel-cli`) ships as a stand-alone open showcase repository.

VulnAdvisor Pro

LOCAL VULNERABILITY TRIAGE & ADVISORY ENGINE

A local engine that ingests findings from Semgrep, Nuclei, and CodeQL and re-ranks them by **real exploitability**, not theoretical severity. Saves security teams days of noise triage per week.

CHAPTER 10 · GAMING UNIVERSE

10 — Consumer Gaming Platforms

Six shipped consumer products serving the Arabic gaming audience:

Golden Arena 518

AI-POWERED FANTASY BATTLE PLATFORM

A fantasy arena that pits characters from anime, video games, and film. AI analyses each character's abilities; an ELO system ranks them; an interactive tier list surfaces the meta.

URL: 61465.github.io/game-zone-golden

Game Zone Archive

ENCYCLOPAEDIA + AI DNA ANALYZER

200 characters × 200 games × 60+ franchises. An AI "DNA" analyser produces personalised game recommendations.

URL: 61465.github.io/game-zone-archive

The Last Save

INTERACTIVE NARRATIVE GAME

An interactive narrative experience about memory and identity. Every decision alters the ending.

URL: 61465.github.io/game-zone-game

GZ Chat Pro

ARABIC-FIRST GAMER COMMUNITY PLATFORM

Voice rooms, embedded mini-games, rank & XP system — an Arabic-first alternative to Discord for the MENA gamer community.

URL: 61465.github.io/GAME-ZONE-CHAT

Waraq Reader

EDUCATIONAL TRANSLATION PLATFORM

Multi-language reading with four study modes and Arabic AI translation support. Positioned as an educational reader for MENA students.

URL: 61465.github.io/read

Golden Arena Configurator

DECISION-ANALYSIS TOOL

Compares games and hardware using radar charts and structured text analysis.

URL: 61465.github.io/game-zone-CONFIGURATOR

CHAPTER 11 · SPECIALISED VERTICALS

11 — Specialised Products

Zwaya Fit v0.3.0

WHATSAPP AI HEALTH COACH

A full AI health coach delivered inside WhatsApp. Baileys + Groq stack. 16 REST endpoints. Groq-Vision meal analysis. Auto-generated weekly weight-tracking PNGs. Eleven achievement badges. Six context-aware alert types on a 10 AM / 6 PM scheduler. Exercise library with TTS audio. Family mode. Forty-feature admin dashboard.

GZ CyberRange v0.1

FINTECH TRAINING LAB · 30 PLANTED VULNERABILITIES

A cyber range with thirty planted vulnerabilities spanning OWASP Top-10 · fintech-specific · Next.js SSR · API misc. Coverage validated at **86%** by an automated NEXUS attack loop. Positioned for corporate security-team training.

Restaurant Management Suite

NEXT.JS + PRISMA + VERCEL

End-to-end restaurant operations. Menu with 200+ items. Deployed for one MENA operator. Prisma schema ready for multi-tenant rollout.

V24 Operations PWA

TALABAT + INSTASHOP DASHBOARD

Multi-channel operations dashboard for MENA food-delivery brands. Shift KPIs. Voice dictation. Supabase sync.

CHAPTER 12 · PUBLISHED RESEARCH

12 — Research Portfolio

SSRN 6870178 · MIT Licence

Ariadne — Adaptive HITL Defence for AI Agents

A four-pillar defence framework against indirect prompt injection targeting autonomous AI agents: (1) least-privilege scoping, (2) tamper-evident observability, (3) synthetic deception, (4) human-gated remediation.

REPRODUCIBLE RESULT

8,000-session AgentDojo benchmark. **90.6% adversarial containment** at 5.9% false-positive rate. Least-privilege pillar alone reaches 56.5% containment at **0% FPR**.

SSRN 6602478 · IEEE TIFS extended bundle in preparation

Oblivion Gate — Zero-Trace Fragmentation in Distributed Data Architectures

Three novel cryptographic protocols (BDSP · STCHP · HS-CPP) for zero-trace fragmentation of sensitive data plus decoy-based threat saturation. The core innovation: instead of encrypting the data, shatter it into individually meaningless fragments, distribute them, and populate the space with plausible decoys that saturate any adversary's search budget.

arXiv-ready · Nexus Zone Commercial 1.0

LucidCode — Program Verification with Grounded Anthropomorphic Interfaces

The academic paper behind LucidCode. Unifies compositional reasoning, adversarial mutation testing, provenance tracking, citation enforcement, and a zero-LLM truth path. Central philosophical claim: *you do not need a language model to know the truth — you only need one to describe it.*

CHAPTER 13 · COMPETITIVE RESULTS

13 — Kaggle Track Record

9,753 teams competed across four active Kaggle tracks. Independent submissions against university and corporate teams. Verified live scores:

91.755

JED (Top 14.8%)

0.936

RSNA AUC

1.78

ARC-AGI-3

30.14

ARC-AGI-2

THE DISTILLED PLAYBOOK

PLAYBOOK_KAGGLE.MD — VERBATIM-FIRST

Extracted from twelve failed modification attempts across three competitions: **clone the strongest public notebook verbatim first, tune one knob per submission, never combine changes**. Eleven of twelve exploratory modifications regressed. The playbook is now a shared internal artefact for future Kaggle work.

PUBLISHED WRITEUPS

- **ARC-AGI-2 Paper Track:** "The Symbolic Ceiling on ARC-AGI-2" — a novel application of Friston's Free Energy Principle to ARC; $p(\tau)$ computed on 320 tasks.
- **Pokémon TCG Strategy:** An α -Rank statistical study with Wilson confidence intervals on 200 games per cell, plus a five-signature strategic framework.

CHAPTER 14 · RESPONSIBLE DISCLOSURE

14 — Applied-Security Track Record

Nine engagements across HackerOne, Bugcrowd, and internal ranges. Two accepted. Several under active triage.

PROGRAM	PLATFORM	SEVERITY	VULNERABILITY	STATUS
Syfe	HackerOne #3914035	CVSS 8.6	Web Cache Deception (CWE-524)	Accepted
Gearset	Bugcrowd 0e7d2c90	P3	Excessive GitHub-OAuth scope (repo + write:repo_hook) on staging.claytonapp.com	Accepted
Shopify	HackerOne #3919475	CVSS 8.7	Unified 7-vector chain (UCP SSRF + GCP trace leak + OTLP unauth + tenant spoof + schema oracle)	Triage
AXIS OS	Bugcrowd 8a7425f0	P2	ACAP D-Bus wildcard bypass of CVE-2025-5452 (firmware-validated on M1075-L 12.11.77)	Submitted
eToro	Bugcrowd f83be6be	P1	1-click ATO on Delta Desktop (6 chained bugs, live PoC)	Submitted
Okta AtSpoke	Bugcrowd 6d26be3a	P1	Application-wide CSRF via hardcoded XSRF-TOKEN=JustAskSpoke!	Submitted
LaunchDarkly	Bugcrowd a91d8daa	P2	IAM read-bypass — any api-* token enumerates 16 account endpoints including member PII regardless of no_access + deny	Submitted
Magic Labs	Bugcrowd 696307c8	P3	reCAPTCHA bypass + user enumeration (13 staff accounts confirmed, including CEO and CTO)	Submitted
GZ CyberRange (self-built lab)	Internal	—	30-vulnerability fintech training lab; 86% hunter coverage validated	Live

CHAPTER 15 · ROADMAP AHEAD

15 — Roadmap

NEXT 90 DAYS

- Onboard the first three paying pilots for **Laundry** at USD 499 / month per repository.
- Publish first hospital pilot for **midcine** in Egypt.
- Complete peer-review preparation for the **LucidCode** paper.
- Announce the Arabic-first LLM (Nexus Zone AI model — evaluation phase).

NEXT 6 MONTHS

- Hosted multi-tenant dashboard for **Laundry**.
- SOC 2 certification path opened to unlock enterprise sales.
- Ten paying customers · USD 30K ARR target.
- Publish the **Ariadne** IEEE TIFS extended bundle.

NEXT 12 MONTHS

- Nexus Zone at USD 100K ARR.
- First pre-seed or seed round.
- Team expansion to three engineers.
- ARIA v10 with open-source companion infrastructure.

“

A product is not judged by the demo. It is judged by the reproducible run.

Engage With Nexus Zone

Every product in this catalog is real, published, and independently verifiable. Requests for evaluation copies, pilot programmes, licensing conversations, or research collaboration are welcomed.

CONTACT

abdarahman10555@gmail.com

Response within one business day.

Nexus Zone — *Where technology meets craft.*

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